Welcome!

Have you ever felt like your training wasn't producing results?

If yes! Meet Apex

Your AI-powered personal trainer

Apex, maximizing your training

- **Aaron Ricca**
  - Electrical and computer engineering
  - Ex professional skier
  - Climber
- **Derya Özsoy**
  - Informatik
  - Kickboxing

What's the problem ?

- training is far from optimized
- hard to find good exercises
- time is always too little

How does Apex work?

- the End User enters his **data**
- Apex learns with the collected **User data** and **data** from Internet and tries to find common patterns.
- At the end the User receives a personalized training plan

Data time !

Input attributes:

- Objective
- Available time
- Favorite exercises
- Biological attributes (Height, body fat%, medical condition, etc.)
- Alimentation (Cuisine, Diet type, enhancing substances, etc.)

Output attributes:

- Training periodization
- List of exercises
- Alimentation improvement tips
- Corresponding improvement % [Continuous]

Data Sources 📊

Research & Literature 📚

- Studies & publications
- Scientific papers

Real-World Data 💪

- Gyms & fitness centers
- Testers & athletes

User-Generated 📱

- Surveys & questionnaires
- Smartwatch data

Continuous Growth 🔄

- Data collected from users

Data Preprocessing 🔧

- Convert different units & scales to SI units
    - Height (feet → cm), Weight (lb → kg), etc.
- **Standardization** → for classification algorithms (KNN, Random Forest)
- **Normalization** → for regression algorithms

Few data?

→ More data?

🧪 Synthetic data!

SMOTE

***S**ynthetic **M**inority **O**versampling **Te**chnique*

- Used for **imbalanced datasets**
  → e.g., 90% fit people and 10% overweight
- Problem: ML models **bias the majority class**
- SMOTE **creates new synthetic data** of the minority class

How SMOTE Works

Our scenario: 10% overweight, 90% fit

-**User A:** Age 35, Weight 195kg, Beginner level

-**User B:** Age 40, Weight 200kg, Beginner level

SMOTE finds two **similar overweight users** and creates a new realistic profile:

-**Synthetic user C:** Age 37, Weight 197kg, Beginner level ✨

✅ Makes the minority class **denser** 10% → 27%

What tools are used for Apex?

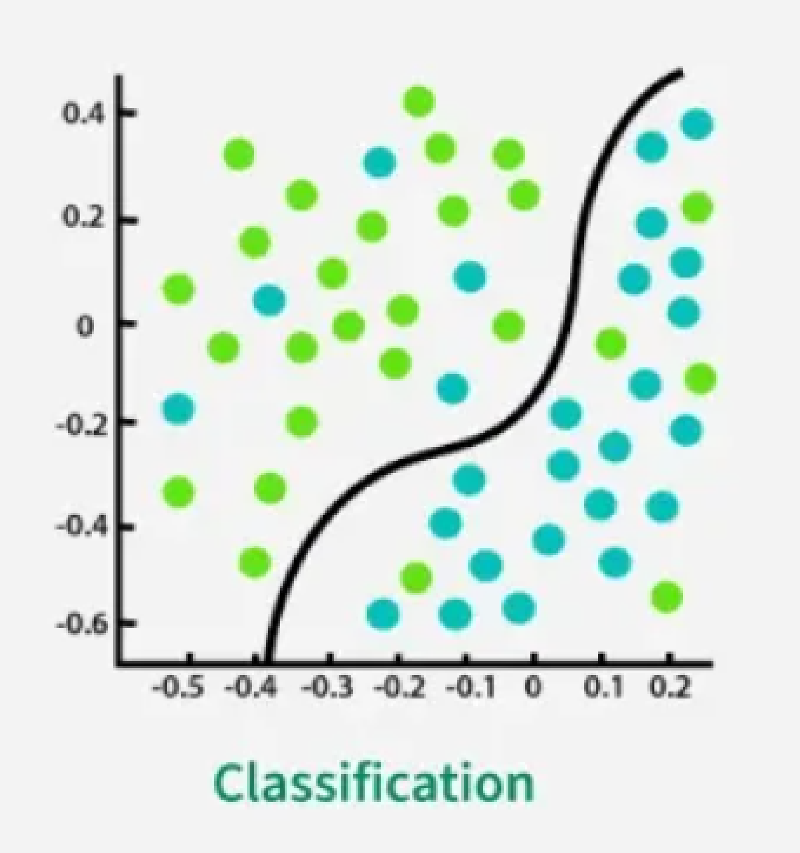
Data Processing 🔧

- **SMOTE** - Balance underrepresented groups
- **Cross-Validation** - Ensure model generalization
- **PCA (Principal component analysis)** - Reduce the data dimensionality

Supervised Learning🔧

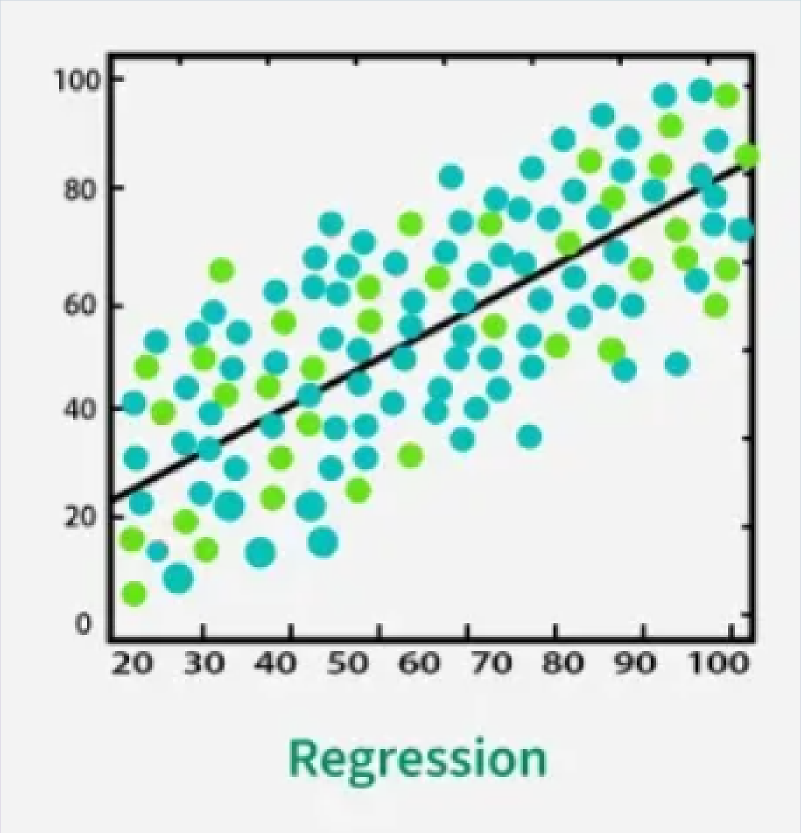
Classification

- predict an outcome or event
- outcome can take two possible values



Regression

- predict continuous values



Association rules

used to discover interesting relationships and dependencies between items in large datasets.
The rules are expressed as "if-then" statements, such as $X \rightarrow Y$

Algorithms 📈

Linear Regression

- Prediction of continuous values
- Uses a linear relationship between input and target variables.
- Corresponds to the "line of best fit."

Logistic Regression

- The target variable is binary
- Uses the logit function to predict probabilities.
- Ideal for classification problems.

Decision Tree

- Makes decisions using a tree-like structure.
- Each leaf represents a decision or prediction.
- Easy to interpret and visually understandable.

K Nearest Neighbor

- Categorizes data based on proximity to other points.
- The new point is assigned to the class of its nearest neighbors.
- Simple, but computationally intensive for large data sets.

Random Forest

- Combination of many decision trees (ensemble method).
- Provides more accurate and stable results.
- Can be used for classification and regression.

Naive Bayes

- Based on Bayes' theorem (probabilities).
- Assumption: Features are independent (naive).
- Fast, efficient, particularly useful for text classification.

Why is Apex better?

- **Open source** - Transparent algorithms, community improvements
- **Data-driven optimization** - Personalized vs generic programs
- **Lower injury risk** - Training plan based on personal condition
- **First of its kind** - Novel ML-based personal training approach
- **Continuous learning** - Improves with more user data

Possible risks, barriers and obstacles

- **Incomplete data can lead to incorrect results.**
- **Time and resources.**
- **Data breach.**

Ethical Considerations

- **Privacy & GDPR compliance** - Sensitive biometric and health data
- **Labor market impact** - Potential displacement of personal trainers
- **Economic effects** - Reduced gym memberships
- **Algorithmic bias** - ML skewed toward young, fit persons
  - Risk: Inappropriate recommendations for underrepresented groups
  - Can lead to injuries in elderly, etc.
- **Dual-use concerns** - Technology repurposed beyond original intent
  - Example: Military training

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Apex: Any questions?

We're happy to dive deeper.